

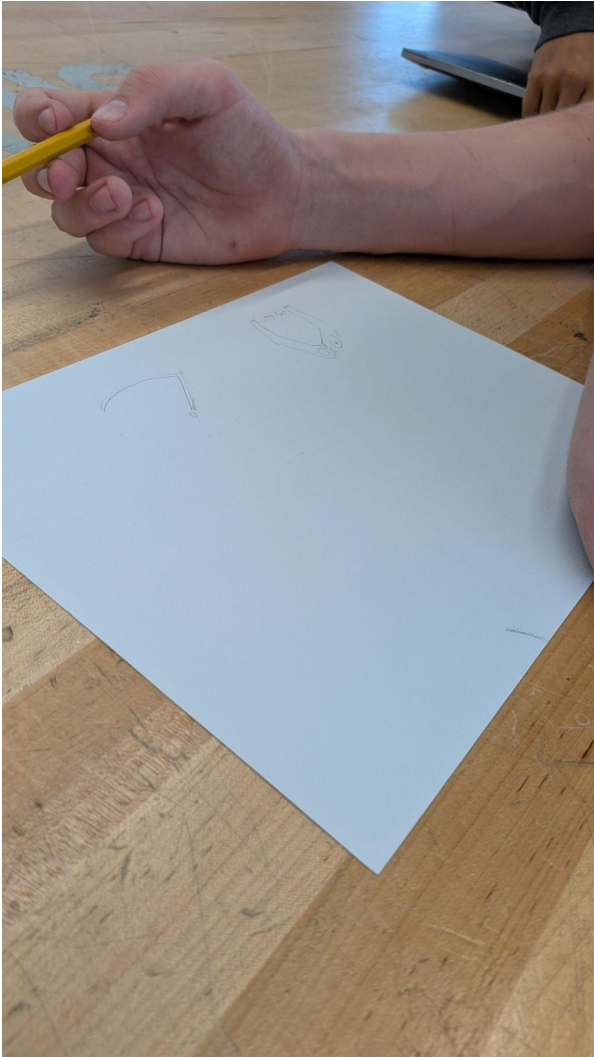
Name: Arnab Krishnan

Report Date: Week of September 8th

Work Completion Conflicts: Labor Day– No Monday. As a result, we lacked much-needed time for prototyping, resulting in doing parts of it (namely onshape) at home.

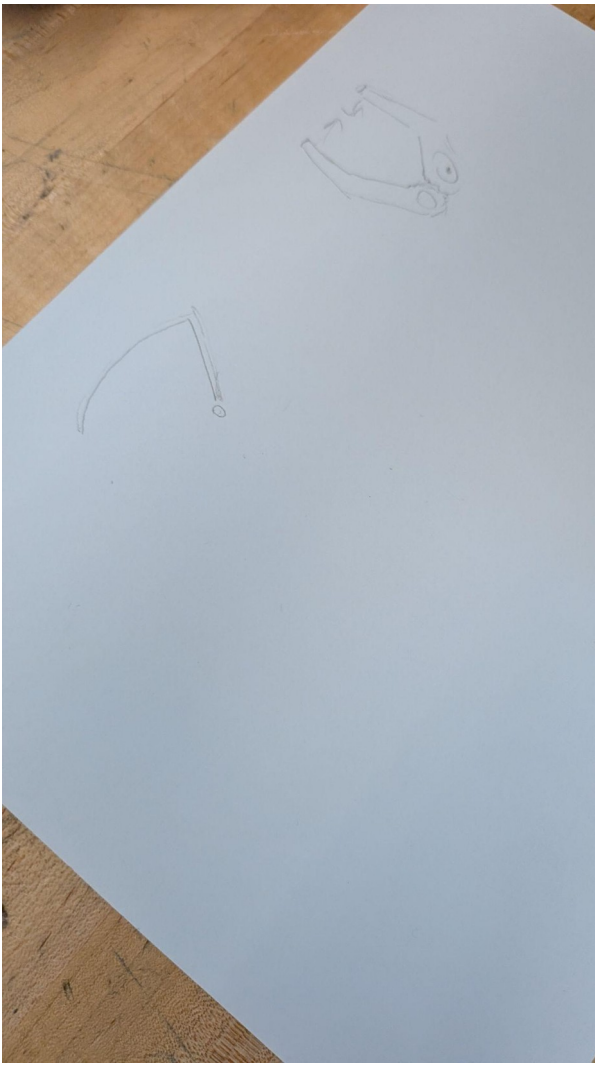
Completed Tasks

Brainstorming/Designing Prototype - Wednesday - 45 minutes



The goal for this week was to prototype / design a specific mechanism that could reliably grab, and release, an irregular object; in our case, this was a white PVC cross fitting with five open ends. The main issue, and what makes this piece tricky to grab, is that it has no flat faces to grab. Its open holes on each side are tricky, although they can be exploited for hooks. Most challenging, though, is that depending on how it lands, it can present an entirely different profile to the mechanism we are designing. When laying such that there is just one connection pointing upwards, it has a cross-shaped cross section with respect to the table. However, it can also be on its side, presenting a T-shaped cross section. Worst of all, it can be resting diagonally, presenting no uniform cross section, and no intuitive way to be picked up by a mechanism.

In this case, my individual contribution was contributing to the ideation, and generating initial concept sketches. I started with a hook style arm idea, because my first instinct was that a hook could catch one of the cross openings and lift it. After thinking through it more, I realized a hook would only work if the cross landed with an opening facing sideways, and this was far too chance-dependent for it to be a reliable system. I also designed a two-arm pinching idea, however, we ultimately decided that it was a bit too convoluted, and there was an easier solution...



The idea we eventually settled on was to use a rod of soft, compressive foam wheels. Since this was just a prototype, they would be spun by drills. Two wheels, spinning inward towards each other, would squeeze the PVC cross between them and pull it in, similar to how an intake on a competition robot works.

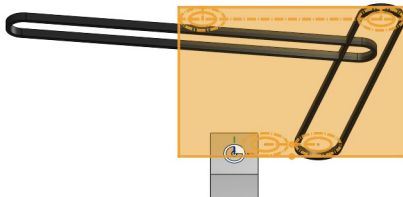
Our reasoning for how it would work is that soft wheels would deform around the pipe, since the PVC pipe presented rounded surfaced, and the spinning motion in particular would do the work of pulling it in.

Our biggest challenge in this session was time. Because of Labor Day, we lost 50 minutes on Monday, so we only had 45 minutes for brainstorming on Wednesday, before we had to move on to CAD, which is described in the next task!



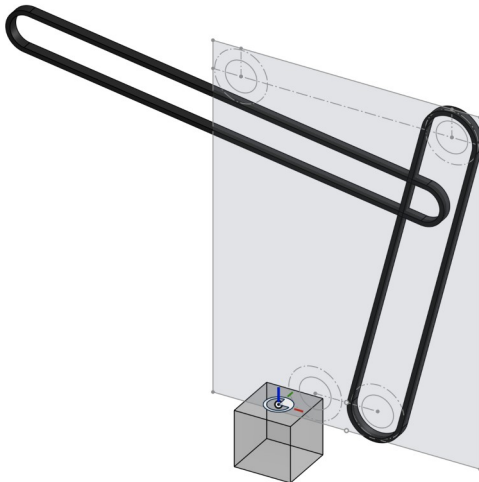
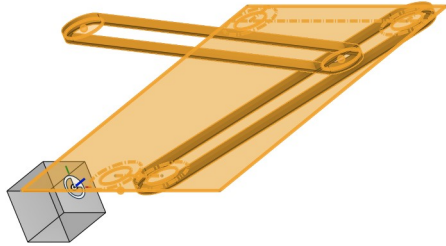
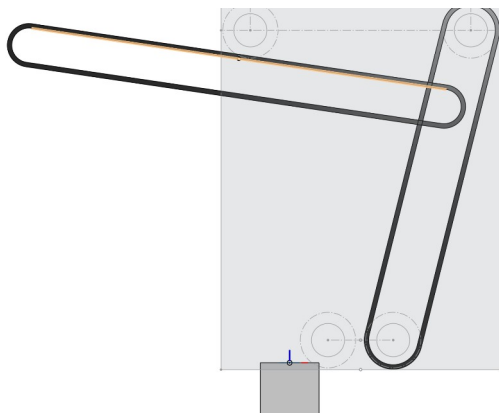


Computer Aided Design (CAD) of the Prototype - Wednesday + Afterschool + beginning of Friday, 1 hour



After brainstorming we moved onto Onshape to model the prototype of the grabbing mechanism we had designed.

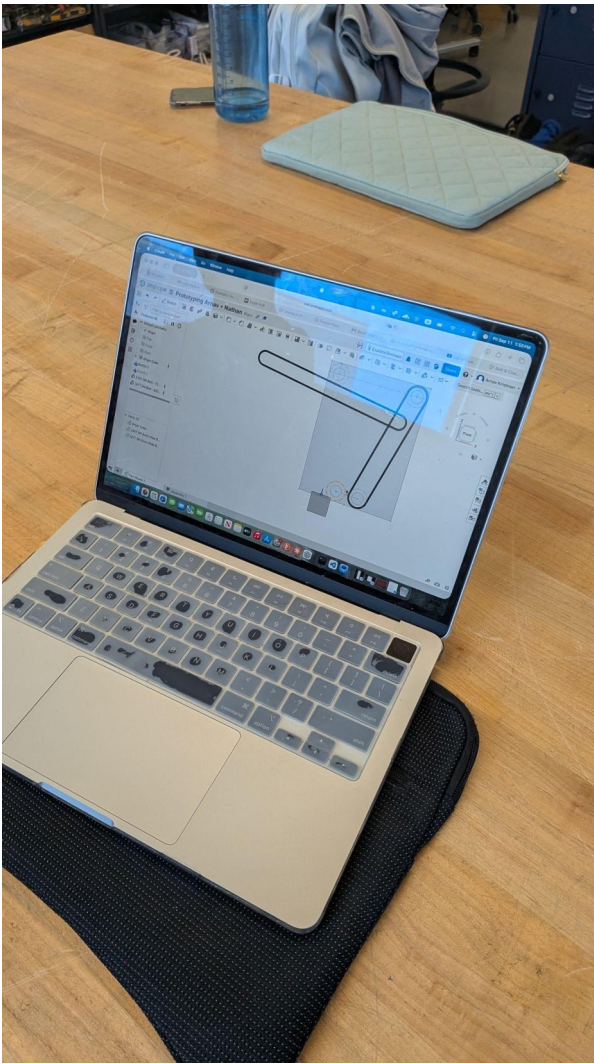
The screenshots show the design in a number of orientations and views. The layout has 4 bearings, with two of them connected to rods that would have the foam wheels, and the other two connected to drills that would be powering the entire mechanism.



The orange highlighted plate in the view is the laser cut side plate that everything would mount onto. We chose to use drills and pulleys rather than putting the drills directly on the wheel shaft because it was awkward to hold two drills at the bottom of the mechanism (we probably would have had to hold them upside-down, for starters, when lowering the mechanism).

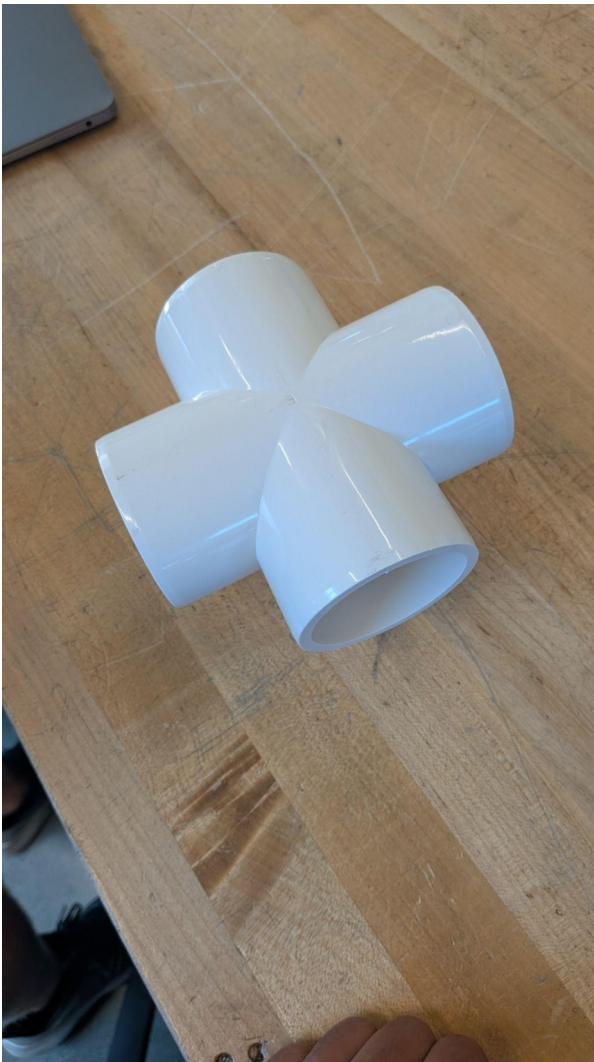
The main challenge we encountered in CAD was getting the belt path and gear sizes to be consistent. Specifically, the belts and gears we had access to (or at least were using) in the shop were not all the same diameter/length, and we did not have them on-hand when modeling, so we had to estimate. In fact, that estimation ended up causing problems in the construction because the belt lengths in the model did not perfectly match what we actually had. However, the mismatched wheels were mismatched in pairs, meaning that they were moving at the same speed (two of the four wheels were the same, and were different from the other two which were also the same, if that makes sense). In hindsight, we should have measured every belt and gear we were planning to use before CADding, as this likely would have saved time later, and also made the process just generally easier.

Construction (laser cutting, etc.) of the Prototype - Friday, 1 hour



Friday was our construction day! We started by laser cutting the side plates from the CAD file. Initially, it was a bit difficult to find used pieces of wood that had enough material to cut our piece; Will, our lead, explained that in the future, smaller side plates probably would have sufficed (although he said he didn't want to correct us initially, but to let us make our own mistake and learn). We mounted the bearings into the holes, and hot glued them in. However, as I will explain later, this was perhaps our most critical mistake, and definitely the most critical in the construction process. We mounted the gears and belts, attached the soft compressive wheels onto the hex shaft, and placed the drills onto the hex shaft. My contribution was assembling one (physical) half of the prototype, meaning 4 bearings, four gears, two belts, one drill, 10 soft compressive wheels, 2 hex shafts, etc.

The result was somewhat limited success. The wheels did spin, and the foam did compress against the sides of the PVC pipe connector. We were able to grab the cross and lift it into our "chamber", and it would stay in there as long as we kept the wheels running. However, there were three main reasons there was limited success. First, the belts and gears were different lengths and sizes, which caused some to have significantly more tension than others. This uneven force load caused some dramatic physical failures, which could happen only for the second reason: we did not put enough hot glue on the bearings. As a result, when the drills began to spin, and more tension was on the belts, occasionally the bearings would fly out. Even worse, once they were out once, the hot glue keeping them in was gone. Due to the limited time, we



had already put the hot glue gun away, so we couldn't add more. We had to get resourceful, using masking tape. Finally, and this is actually something that **Mr. Hine told** us was an issue- in certain orientations of the PVC pipe, we would have to rotate the mechanism, sometimes as much as 90 degrees. Mr. Hine explained that you never want to build a mechanism that requires such a high degree of precision, as it will not be reliable. A good mechanism should be forgiving, ours was not.

Next time I am prototyping, I will remember to (a) re-measure EVERYTHING before I start CADding, to match the actual parts to the design, (b) ensure that everything is actually fastened and secured, to prevent significant physical failures, and (3) ensure I am thinking about the problem from all angles- Literally! The mechanism didn't work at the angle where the PVC pipe was on its side. Of course, this was partly due to losing significant amounts time; we couldn't spend forever on brainstorming.



